



SIMATS ENGINEERING

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Course Code: CSA5115

Slot: A

Course Name: Cryptography and Network Security

Course Faculty: Dr .T.Sangeetha

Project Title: QR-Code based secure authentication system

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description: (here you write what you did in this project (contribution) including Model Description)

The QR-Code Based Secure Authentication System is a secure login solution that uses QR codes to verify and authenticate users. Users first register their details and generate a unique QR code linked to their account. During login, the QR code is scanned and its encrypted information is verified by the system. The system checks the QR code validity, expiry time, and user credentials before granting access. Successful and failed login attempts are recorded in an activity log for monitoring and security. This system reduces the risk of password theft and unauthorized access. Overall, it provides a simple, fast, and secure authentication mechanism using QR-code technology. The system also protects sensitive authentication data through encryption, ensuring that the information stored in the QR code cannot be easily misused. Temporary or expired QR codes can be rejected automatically to prevent replay attacks. The activity log helps administrators track login activities and identify suspicious access attempts. This approach can be useful in banking, educational institutions, offices, healthcare systems, and other applications where secure user authentication is required.

Student Signature

Guide Signature